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ORIGINAL ARTICLE

Maxillary sinus 3D segmentation and reconstruction from cone beam CT data sets

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Abstract
Purpose Segmentation of the maxillary sinuses for three-dimensional (3D) reconstruction, visualization and volumetry is sought using an automated algorithm applied to cone beam computed tomographic (CBCT) data sets.
Materials and Methods Cone beam computed tomography (CBCT) data sets of three subjects aged 9, 17, and 27 were used in 3D segmentation and reconstruction. The maxillary sinuses were obtained by propagation from one start point in the right sinus and one start point in the left sinus to the whole regions of both sinuses. The procedure was based on voxel intensity distributions and common anatomic structures, specifically each middle meatus of the nasal cavity. A program was written in C++ and VTK languages to demonstrate the surface topological shapes of the maxillary sinuses.
Results The developed segmentation algorithm separated maxillary sinuses successfully permitting accurate comparisons. It was robust and efficient. 3D morphological features of the maxillary sinuses were observed from three human subjects.
Conclusions Automated segmentation of maxillary sinuses from CBCT data sets is feasible using the proposed method. This tool might be useful for visualization, pathological diagnosis, and treatment planning of maxillary sinus disorders.

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Keywords Maxillary sinuses · Cone beam computed tomography · Image segmentation · 3D shape reconstruction

Abbreviations

Cone beam computed tomography	CBCT
Three-dimensional	3D
Two dimensional	2D
Millimeter	mm

Introduction

Understanding the three-dimensional(3D) surface morphology of the maxillary sinuses and changes in volume over time can be important in detecting and monitoring pathological conditions affecting the sinuses and determining outcomes of treatment [1–5]. The incidence of sinusitis and other pathoses in the maxillary sinuses is comparatively high, so, analysis of the morphology of the maxillary sinus has clinical value. For endoscopic endonasal surgery, knowledge of the precise anatomy and anatomic variations of the maxillary sinus boundary of the lateral nasal walls can help the surgeon achieve better results [6].

Several previous studies have looked at the visual appearance of maxillary sinuses and estimation of critical parameters related to the paranasal sinuses; however, most used either direct physical measurements or two-dimensional(2D) transmission radiographic images [7–12]. Some researchers have used combinations of axial, coronal and sagittal views or volumes of the head or partial head without specific segmentation of the maxillary sinuses [13,14]. For volume measurement of paranasal sinuses, materials have been injected into cadavers [15,16]. This is not a practical approach for studying living subjects. Use of computed tomography has been without segmentation for automated measurement and 3D display [17]. Relying on human perception of the 3D shape could yield errors of several millimeters and provide inaccurate impressions of surgical anatomy.

Recently, 3D reconstruction of the maxillary sinuses and volume evaluation from such reconstruction has become an alternative. Nevertheless, few papers have reported on the 3D reconstruction of maxillary sinuses [1,5,18,19]. Most of these studies have been performed on animals rather than man.

In 1999, Kawarai et al. [1] performed CT scanning of a Japanese macaque for validation of their method, then followed this with scans of 20 healthy persons. CT anisometric voxel resolution was 3 mm slice collimation and 2 mm pitch. They reconstructed 3D views of the paranasal sinuses using ISG Technologies' software but computed volumes without separation of the nasal passages. The error ratio was about one-third due to the low resolution of the scans and potential inclusion of the nasal passages.

Kim et al. [18] performed 3D reconstruction and simulation of maxillary sinuses from CT images using a commercial V-work program. Details of their segmentation algorithm were not provided. From the observation of the 3D shape, it was determined that they performed measurements using sequences of reformatted 2D images. Such measurements involve human vision and operations which can lead to errors in measurements.

Nishimura, et al. [5] reconstructed the 3D shape of the portion head of platyrrhine monkey with partial paranasal sinus demonstration.

Jun et al. [19] made a thorough quantitative analysis of paranasal sinus volumes in different age groups and found changes in maxillary sinus volume with respect to subject age and sex. Their 3D reconstruction used a surface rendering technique (Vworks; CybeMed, Seoul, Korea). From the reconstructed 3D images, it can be seen that they did not separate the boundary bone enclosing the maxillary sinus and that some portion of the sinus remained hidden. The neighboring CT slice gap was 2.5 mm which limited accuracy of the estimations performed.

In this paper, a novel segmentation algorithm to provide more accurate volume estimations of the maxillary sinuses is presented. The method involved use of cone beam CT (CBCT) image data sets providing isometric voxels.

Study design

Subjects

For this IRB-approved retrospective, three representative anonymous CBCT data sets from three subjects aged 9, 17 and 27 years, respectively, were selected. Fifty other data sets were also examined to determine common geometrical features of the maxillary sinuses, including their openings to the nasal passage. The data sets had been acquired for various purposes unrelated to the paranasal sinuses but included the whole region of the maxillary sinuses. A typical coronal view of the maxillary sinuses with zoomed details of ostial communication sites is illustrated in Fig. 1.

Anatomic features

From the observation of paranasal sinuses for 50 subjects, it was found that the nasal passages generally expanded laterally with increased age. Figure 2 shows coronal views of three typical maxillary sinuses in individuals aged 9, 17, and 27 years. Each maxillary sinus communicates with the middle meatus of the nasal cavity through a narrow ostium. The opening is a small region compared to the whole sinus. Figure 3 shows the axial views of the three typical paranasal sinuses previously shown in Fig. 2. It can be seen that the opening sites in the axial views appeared larger than those viewed coronally. Thus, segmentation employing the coronal views was preferred.

Methods

The CBCT system (iCAT, Imaging Sciences International, Hatfield, PA) provided high quality images little

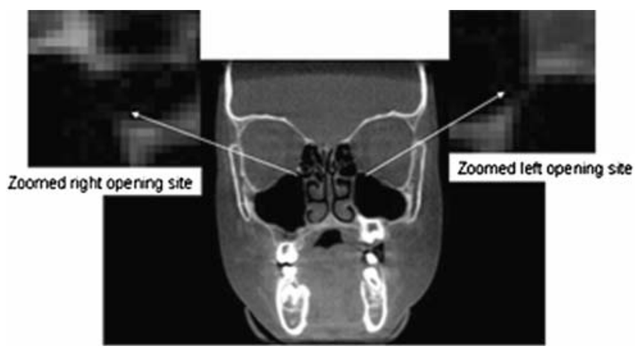

Fig. 1 A typical coronal view of maxillary sinuses with zoomed ostia at *upper left* and *right corners*

Fig. 2 Three typical coronal views of maxillary sinuses for subjects aged 9, 17, and 27 years showing the ostia and the intensity distinction of the maxillary sinuses from the neighboring tissues

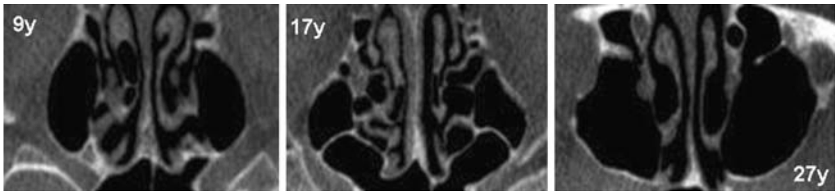

Fig. 3 Three typical axial views of maxillary sinuses for subjects aged 9, 17, and 27 years

noise. Images with 0.4 mm isometric voxels were used. The intensity value of the paranasal sinus contents was zero for air, and ≤5 for any fluid exudate. Such intensities are much smaller than neighboring soft tissues which usually had the intensity range of 40–120, and hard tissues which had the intensity range of 140–255 (Fig. 2).

DICOM 16-bit images were reduced to 8-bit JPEG for the purpose of developing the segmentation algorithm. In our formulation of the segmentation algorithm, a coordinate system was selected so that the "length" of coronal slices was the *x*-axis with the positive direction pointing to the right. The "width" was the *y*-axis with the positive direction pointing downwards. The "height" of the stack of slices was the *z*-axis with the positive direction pointing to the largest slice number. Coronal slices were employed due to the narrowness of the ostial projections in these slices, a factor which reduced computation and improved the accuracy of the segmentation algorithm.

The proposed segmentation algorithm is based on the intensity features within the paranasal sinuses, the common geometrical features of paranasal sinuses observed, and the special features of the opening sites of the paranasal sinuses to the nasal passages. To separate paranasal sinuses from the rest of the image, a point was first selected as the designated starting (reference) point within each maxillary sinus. Since the intensity values in the paranasal sinuses were all equal to zero or ≤5 compared with intensity values above 20 of the neighboring tissues. Hence, the threshold for the intensity of paranasal sinuses was set to 5 in the propagation process for segmentation. Such a threshold value can be set higher or lower by users if different intensities result from different scanning settings or systems.

From the proprietary imaging software provided for CBCT image analysis (iCAT®, Imaging Sciences International, Hatfield, PA), the locations of the reference points were readily determined. Starting from the selected reference point in one of the maxillary sinuses, the six immediately neighboring voxels were examined; namely: left, above, right, and below voxels in the same slice as the reference point, and the two nearest points in the immediately adjacent superior and inferior slices. For convenience, a 2D representation has been used to describe the segmentation algorithm for the right maxillary sinus (Fig. 4). The algorithm principle applied to the 3D object is the same as for a 2D object except that the reference had two more neighboring points. The selected reference point "O" in Figure 4 belonged to the right maxillary sinus. After the procedure is completed for the right maxillary sinus, the same procedure is applied to the left maxillary sinus using a different reference starting point.

In determining a point belonging to the right maxillary sinus, two criteria were used for different regions. The stack of coronal slices was split into two different

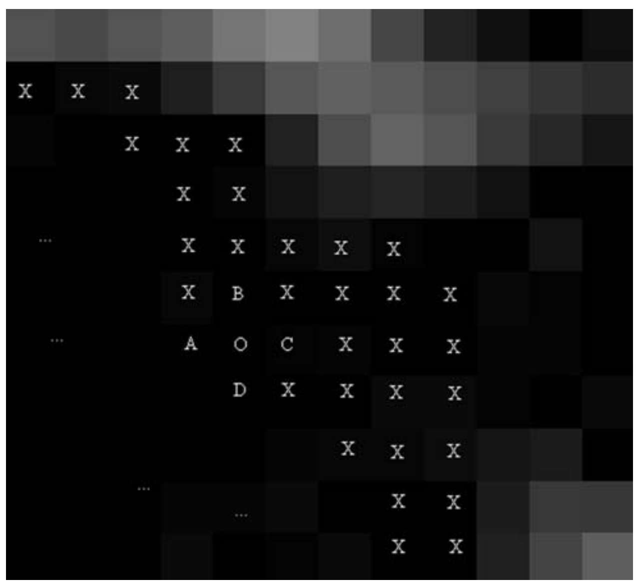

Fig. 4 Illustration of segmentation algorithm on a pixel board locally zoomed from a portion of the right maxillary sinus connecting to the ostium

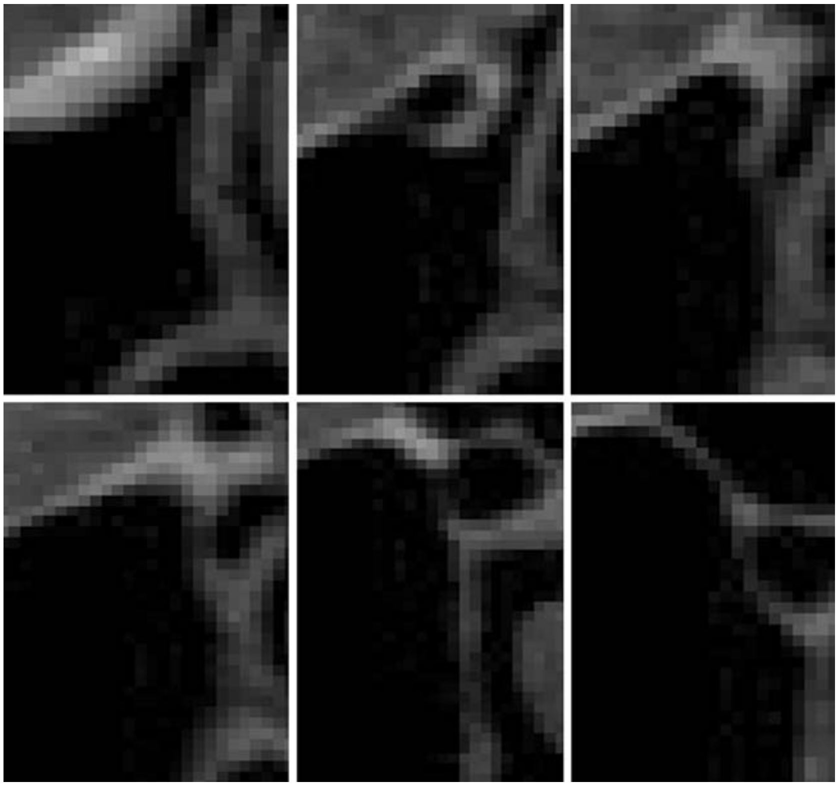

Fig. 5 Six geometric structures around the ostium in coronal views of the right maxillary sinus from the subject aged 9 years to illustrate the effectiveness of the blocking condition required for segmentation

categories, the first category including the coronal slices containing no portion of the opening site and the second category consisting of the rest of the coronal slices. In Fig. 4, point “O” represents the selected reference starting point. “A” (left), “B” (above), “C” (right), and “D” (below) are four immediately neighboring points to the reference point “O”. These points belong to the same category as the reference point “O”. In 3D space, the immediate superior point or immediate inferior point may belong to the second category. For the points in the coronal slices belonging to the first category, they could be identified as belonging to the right maxillary sinus by the set threshold value of 5 only if their intensity values were ≤ 5. For the points in the coronal slices belonging to the second category, besides the intensity threshold, another condition is imposed to block the points inside the nasal passages from being selected as points belonging to the maxillary sinuses. Figure 5 shows several positions of the same maxillary sinus in the stack of coronal slices. It can be seen that any point between the two sides of the narrow ostium can be blocked by imposing a larger threshold of the sum of the distances of the point to the upper and lower sides of the ostium. From the geometrical structure of the opening site projected on the coronal slices in Figure 5 it is apparent that such blocking action is effective in the six slices reflecting the ostium in the coronal views of the subject aged 9 years. Different subjects may have different sizes depending on individual variations; however, their shapes are similar. The ostium is a quite small in area being less than 2% of the surface of the whole sinus. The exclusion of a very small portion of the right maxillary sinus may also be possible due to that the threshold operation for the ostium. From the geometrical structures of the paranasal sinuses, such exclusion is minimal since the maximal distance between the upper and lower sides of the normal ostium is usually less than ten pixels (4 mm).

Four neighboring points “A”, “B”, “C”, and “D” (Fig. 4) were classified based on the applications of the one of the two stated conditions. The points “A”, “B”, “C”, and “D” were classified as points belonging to the right maxillary sinus. It was then necessary to classify the points neighboring point “A”. Point “O” was one of the neighboring points but was already selected and hence did not need re-verification. The examination order for the neighboring points of “A” was clockwise from its neighboring left point, to its neighboring superior point, to its neighboring inferior point. This process continued until a point was encountered that was verified as belonging to the right maxillary sinus for which neighboring points were already selected as belonging – or not belonging – to the right maxillary sinus. Figure 4 shows the selected points (marked) in the right maxillary sinus connected to its opening site. It shows that the procedures based on the two conditions can separate the right maxillary sinus from the rest of tissues successfully by setting the intensity of the right maxillary sinus to 255, corresponding to positions in template images having zero intensity. After the segmentation of the right maxillary sinus, the same procedures can be applied for the left maxillary sinus to complete the segmentation of the maxillary sinuses. The two segmented maxillary sinuses are in the same positions as they were originally, and can be rotated in combination to any orientation.

Results

The proposed segmentation algorithm was combined with VTK visualization generated 3D shapes of the selected subjects. The segmentation algorithm of the maxillary sinuses separated the paranasal sinuses completely from all neighboring tissues so that the paranasal sinuses could be seen from any perspective.

Figure 6 shows the anterior and superior views of maxillary sinuses of the three subjects. The right and left maxillary sinuses are quite symmetric about the vertical plane equidistant to both sinuses but with slight difference due to individual shape differences. The right and left maxillary sinuses appeared more symmetrical from the 3D reconstructions than from the 2D slices (axial, sagittal, and coronal). From this point, the 3D shape reconstruction gives better understanding of the real anatomic structures. Non-symmetry of the right and left maxillary sinuses was reflected in volume differences.

Figure 7 shows the top and bottom views of the maxillary sinuses of the three selected subjects. The surface shapes of these views are roughly triangular. Figure 8 shows the right and left views of maxillary sinuses of the three subjects. The same observations from Fig. 6 about the symmetry and sizes of the right and left maxillary sinuses can also be observed from the top, bottom, and lateral views in Figs. 7 and 8. A general conclusion can be made that the right and left maxillary sinuses of the selected subjects were basically symmetric with respect to an imaginary vertical plane equidistant to both sinuses.

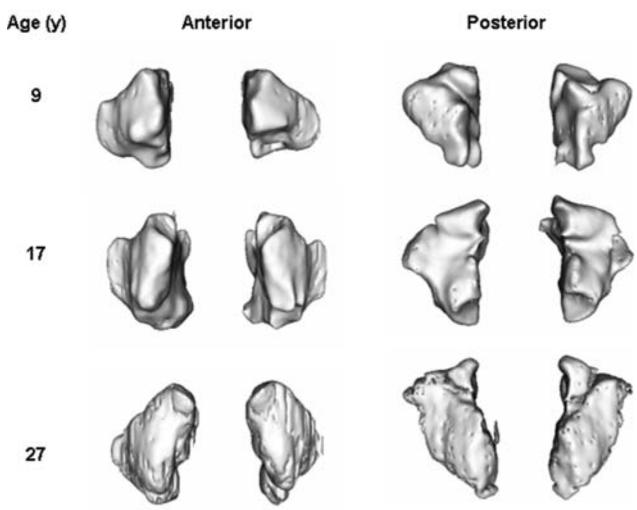

Fig. 6 Anterior and posterior views of the maxillary sinus 3D reconstructions for subjects aged 9, 17 and 27 years

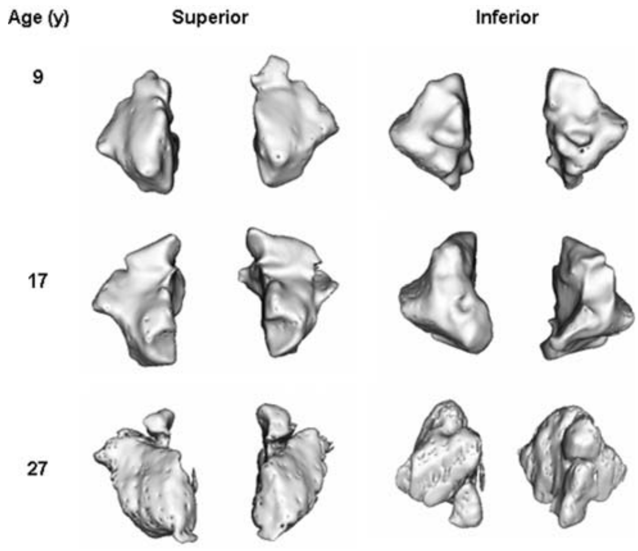

Fig. 7 *Top* and *bottom plan views* of the maxillary sinus 3D reconstructions for subjects aged 9, 17 and 27 years

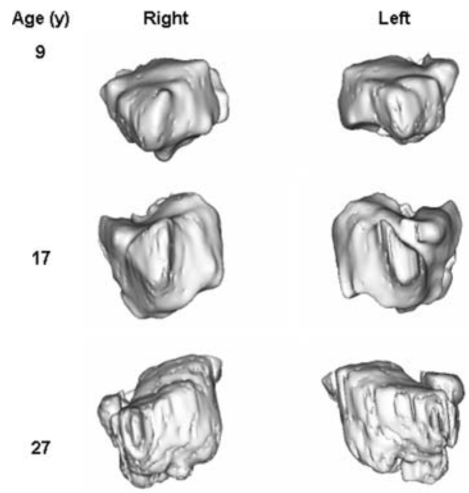

Fig. 8 *Right* and *left lateral views* for of the maxillary sinuse 3D reconstructions for subjects aged 9, 17 and 27 years

For all three selected subjects, the intensity thresholds could be set to the value 5, and this also proved to be applicable to all 50 data sets studied for the initial anatomic structure observation. So, the intensity thresh-

Table 1 Volumes of the right and left maxillary sinuses for selected subjects

Max. Sinus	Volume (mm^3) 9 years	Volume (mm^3) 17 years	Volume (mm^3) 27 years
Right	796243	1856941	1279725
Left	701632	1683821	1195962

old was not sensitive to variations in the scanning environment or subject factors. The ostial vertical gap for the 50 humans studied was in the range of 0.3–6 mm. The threshold value of the vertical gap lengths of the ostium can be found by trial and error. The smaller of this threshold the better the segmentation results since the low threshold can keep the original shape in this high curvature region.

Volume estimation for the maxillary sinuses was performed simultaneously during the segmentation process. The numbers of voxels classified to the right and the left maxillary sinuses are counted separately in the classification process. Table 1 shows the volumes of the right and left maxillary sinuses for the selected subjects aged 9, 17 and 27 years. The volume of the right maxillary sinus is larger than that of the left maxillary sinus for the three selected subjects.

Discussion

The selection of the seed point was less than a minute, and often just a matter of seconds, depending on how many slices there were in the data set based on sinus dimensions. Hence, the total time involved using the proposed segmentation method was one to two minutes. This is a great saving of time compared to the approximate three hours needed for manual segmentation of maxillary sinuses from more than 300 axial slices. The consistent finding of the larger sinus being on the right side in our selected subjects fits well with previously reported data [1,14,19]. The estimated right and left maxillary volumes also matched well with the reported data corresponding to the age groups [1,14,19]. Trial and error will probably be needed to calibrate for accurate segmentation of the ostial communication site as the intensity threshold might vary for different scanning environments. The high quality scanner used in the study (iCAT, Imaging Sciences International, Hatfield, PA) generated a relatively uniform intensity for the air filled maxillary sinuses. Since the resolution of CBCT images is quite high (selected in the range of 0.2-0.4 mm for the system utilized), the accuracy of the proposed automatic segmentation algorithm can be within 0.4-0.8 mm. Due to the imprecision of human vision error using manual segmentation can be greater. Much time is needed for the manual segmentation operation where increased precision is desired. The proposed segmentation algorithm considers the specific anatomic structure and therefore can be used only for the maxillary sinuses. However, it could also be applied to segmentation of other anatomic organs if modified to accommodate the corresponding specific geometrical structures. The main theme of the algorithm remains unchanged. Therefore, besides further study of the geometrical changes of the maxillary sinuses, other clinical applications might be considered in the future for the proposed segmentation algorithm.

Conclusions

The proposed segmentation algorithm separated the maxillary sinuses completely from the neighboring tissues. Segmentation was performed simultaneously for the right and left maxillary sinuses; hence, the relative orientation and position of the two sinuses is retained. Translation and rotation of one maxillary sinus result in the contralateral maxillary sinus translating and rotating in the same manner. The 3D morphologies can be seen from any perspective and the volumes of both maxillary sinuses can be accurately estimated. The proposed algorithm is robust and efficient.

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